Diffractive ϕ Meson Electroproduction at CLAS12: Monte Carlo Tuning with Neural Network Fast Simulation

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Abstract

We present results of Monte Carlo generator tuning for diffractive ϕ meson electroproduction at CLAS12. The cross-section model parameters are determined by fitting acceptance-corrected experimental yields in bins of Q^2, $|t|$, and W. Five CLAS12 datasets (RGA Fall 2018 inbending/outbending, RGA Spring 2018 inbending/outbending, RGA Spring 2019 inbending) are analyzed using a neural network-based fast Monte Carlo simulation for detector acceptance.

Contents

1 Cross-Section Model

The differential cross section for exclusive ϕ meson electroproduction $ep \to ep\phi$, $\phi \to K^+K^-$ is parametrized as:

$$\frac{d\sigma}{dt} = \frac{d\sigma_T}{dt} + \varepsilon\,\frac{d\sigma_L}{dt} \tag{1}$$

where ε is the virtual photon polarization parameter.

1.1 Transverse Cross Section

The transverse cross section is:

$$\frac{d\sigma_T}{dt} = \sigma_T(W, Q^2) \cdot b_t \cdot e^{b_t \cdot t} \tag{2}$$

with

$$\sigma_T(W, Q^2) = \alpha_1 \left(1 - \frac{W_{\text{th}}^2}{W^2}\right)^{\alpha_2} W^{\alpha_3} \left(1 + \frac{Q^2}{m_\phi^2}\right)^{-\nu_T} \tag{3}$$

where $W_{\text{th}} = M_N + m_\phi = 1.96$ GeV is the production threshold, $m_\phi = 1.019412$ GeV, and $M_N = 0.938272$ GeV.

1.2 Longitudinal Cross Section

The longitudinal cross section is related to the transverse via:

$$\frac{d\sigma_L}{dt} = c_R \cdot \frac{Q^2}{m_\phi^2} \cdot \frac{d\sigma_T}{dt} \tag{4}$$

1.3 Model Parameters

The free parameters of the model are:

- α_1 — overall normalization [nb]
- α_2 — threshold exponent: $(1 - W_{\text{th}}^2/W^2)^{\alpha_2}$
- α_3 — W power law: W^{α_3}
- b_t — t-slope [GeV^{-2}]: $b_t \cdot e^{b_t \cdot t}$
- ν_T — Q^2 suppression power: $(1 + Q^2/m_\phi^2)^{-\nu_T}$
- c_R — σ_L/σ_T ratio coefficient (fixed to 1.0)

2 Monte Carlo Generator

The Monte Carlo generator `diffrad_gen_exp` is based on the DIFFRAD code by I. Akushevich (1998) for diffractive vector meson electroproduction with QED radiative corrections in the collinear approximation.

Events are generated in the kinematic space (Q^2, y, t, ϕ_h) using an accept-reject method. The proposal distribution uses an exponential t-slope for efficient sampling. The cross-section model parameters (α_1, α_2, α_3, ν_T, b_t, c_R) are read from the input file, allowing iteration without recompilation.

The generation parameters used for this iteration are:

Parameter	Value	Description
α_2	-1.259	Threshold exponent
α_3	0.697	W power law
b_t	1.313	t-slope [GeV^{-2}]
ν_T	2.306	Q^2 suppression power
c_R	1.000	σ_L/σ_T ratio (fixed)

The event statistics per dataset:

Dataset	E_{beam} [GeV]	N_{gen}	Q^2 range [GeV^2]
Fall18 inb	10.6	10M	[0.5, 7.0]
Fall18 outb	10.6	5M	[0.5, 7.0]
Sp18 inb	10.6	10M	[0.5, 7.0]
Sp18 outb	10.6	5M	[0.5, 7.0]
Sp19 inb	10.2	10M	[0.5, 7.0]

The output is in LUND format with 7 particles per event: beam e^-, target p, scattered e^-, K^+, K^-, recoil p, and (for radiative events) a hard photon.

3 Neural Network Fast Monte Carlo

Full GEANT4-based CLAS12 simulation and reconstruction is computationally expensive. We use a neural network (NN) based fast Monte Carlo that provides:

- Acceptance probability: whether a particle with given (p, θ, ϕ, v_z) would be detected
- Momentum smearing: realistic resolution effects

Separate NN models are trained for each particle species (e^-, p, K^+, K^-) and each dataset configuration (beam energy, torus polarity). An event is accepted if all four final-state particles (e^-, p, K^+, K^-) pass their respective acceptance models.

Typical overall acceptance (all 4 particles detected) is 5–10%, dominated by the K^- acceptance at backward angles.

4 $M(K^+K^-)$ Signal Extraction

The ϕ meson yield $N(\phi)$ is extracted in each kinematic bin by fitting the K^+K^- invariant mass distribution $M(K^+K^-)$ with:

$$f(M) = A \cdot G(M;\, \mu_\phi, \sigma) \;+\; B\,(M - M_{\text{th}})^n\, e^{-bM} \tag{5}$$

where:

- $G(M;\, \mu_\phi, \sigma)$ is a Gaussian with fixed $\mu_\phi = 1019.5$ MeV and fixed $\sigma = 4.7$ MeV
- $M_{\text{th}} = 2m_K = 987.354$ MeV is the K^+K^- threshold
- A, B, n, b are free fit parameters

The signal yield is $N(\phi) = A \cdot \sigma\sqrt{2\pi}/\Delta M_{\text{bin}}$. Binning is performed in Q^2, $|t|$, x_B, and W with 5 bins per variable.

4.1 $M(K^+K^-)$ Fit Results

RGA Fall 2018, inbending

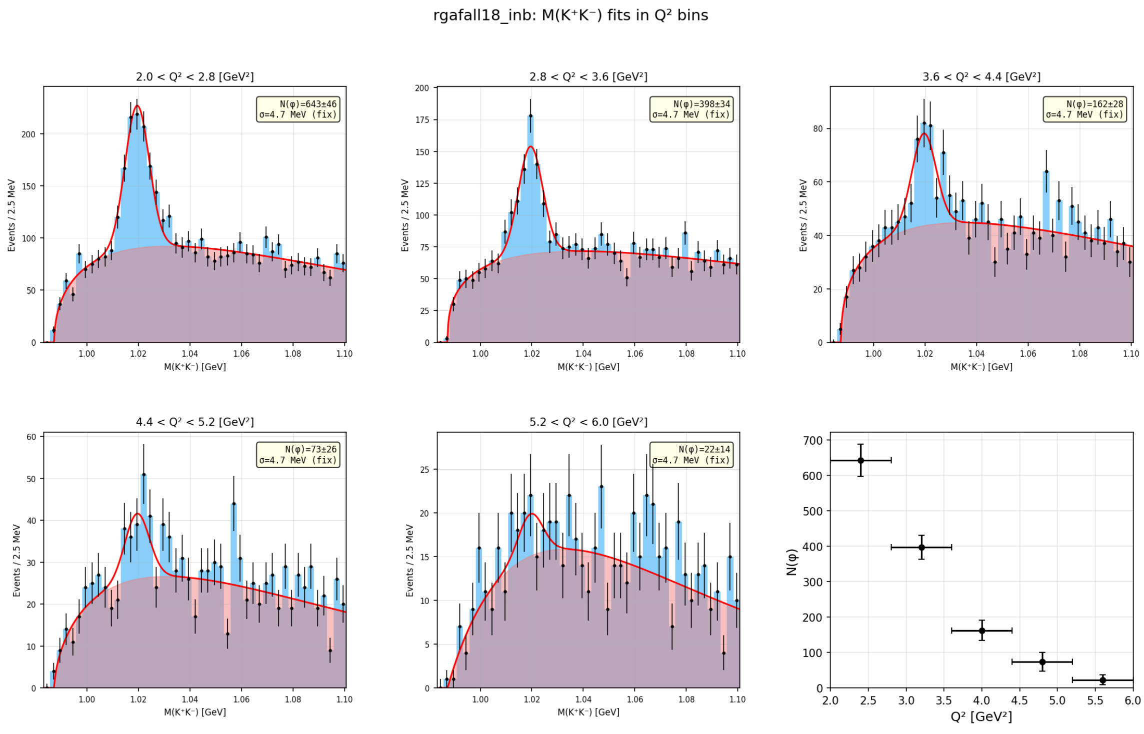

Figure 1: $M(K^+K^-)$ fits in bins of Q^2 for RGA Fall 2018, inbending.

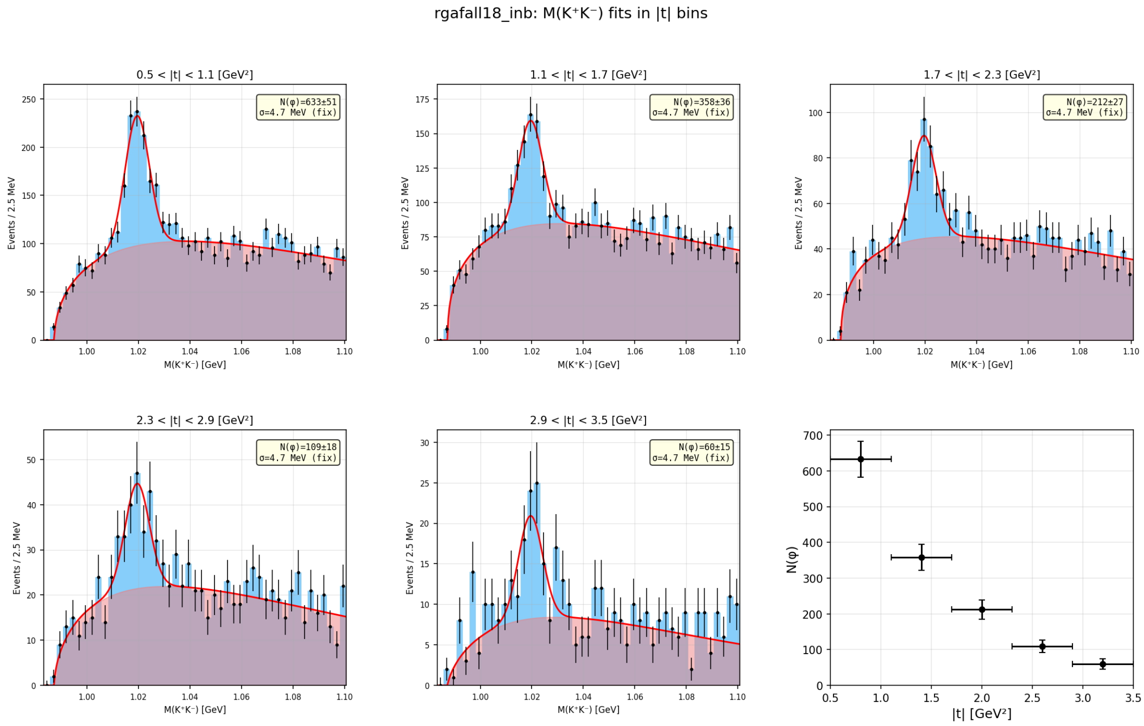

Figure 2: $M(K^+K^-)$ fits in bins of $|t|$ for RGA Fall 2018, inbending.

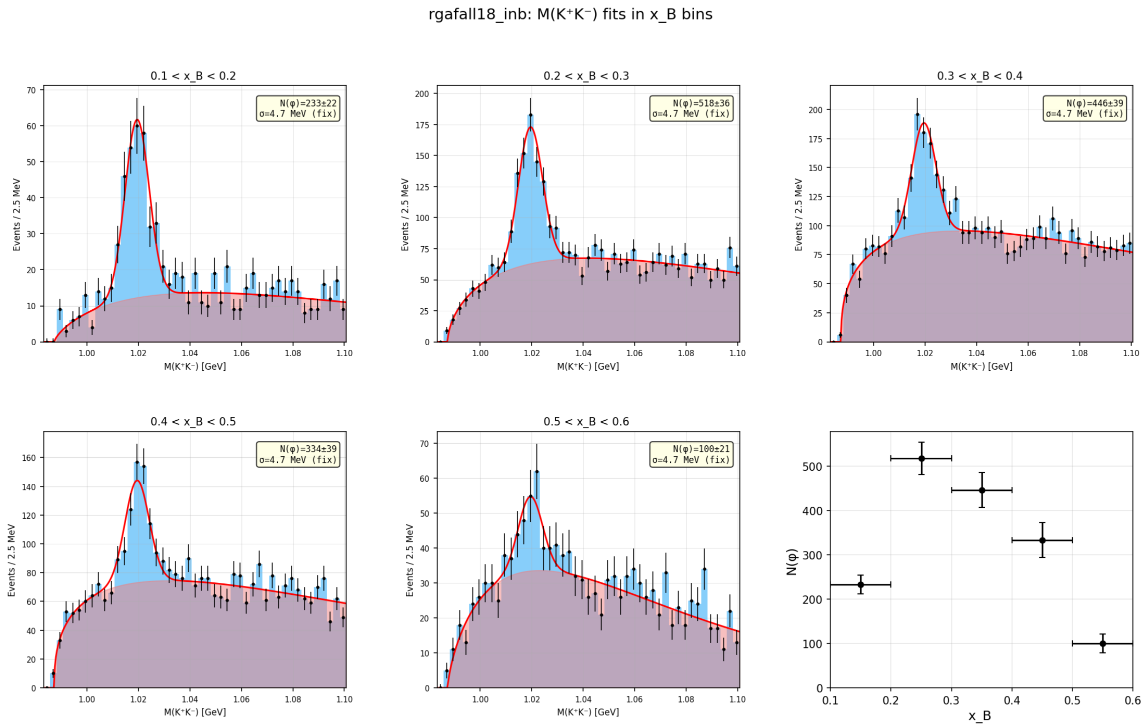

Figure 3: $M(K^+K^-)$ fits in bins of x_B for RGA Fall 2018, inbending.

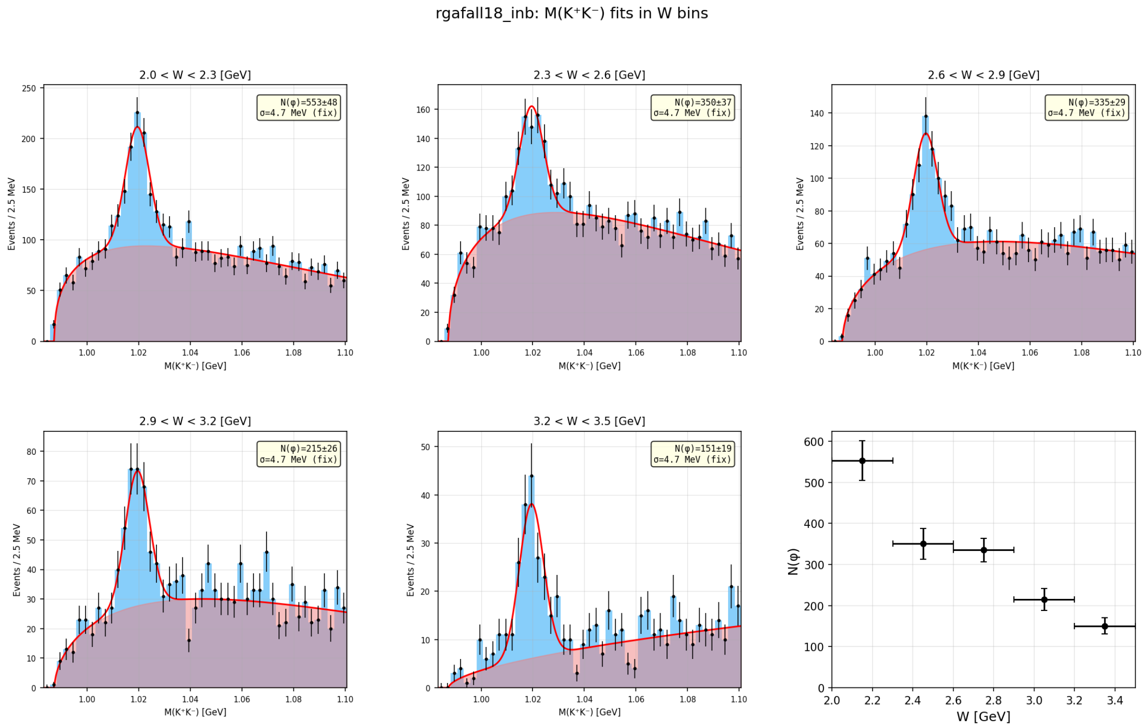

Figure 4: $M(K^+K^-)$ fits in bins of W for RGA Fall 2018, inbending.

RGA Fall 2018, outbending

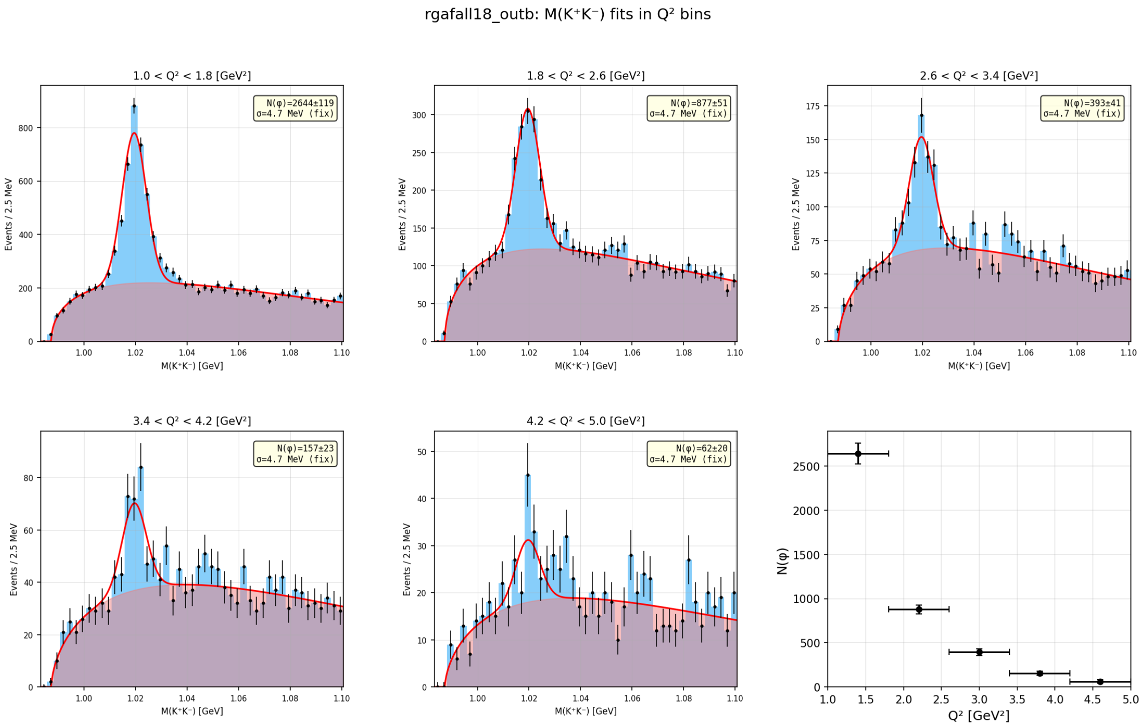

Figure 5: $M(K^+K^-)$ fits in bins of Q^2 for RGA Fall 2018, outbending.

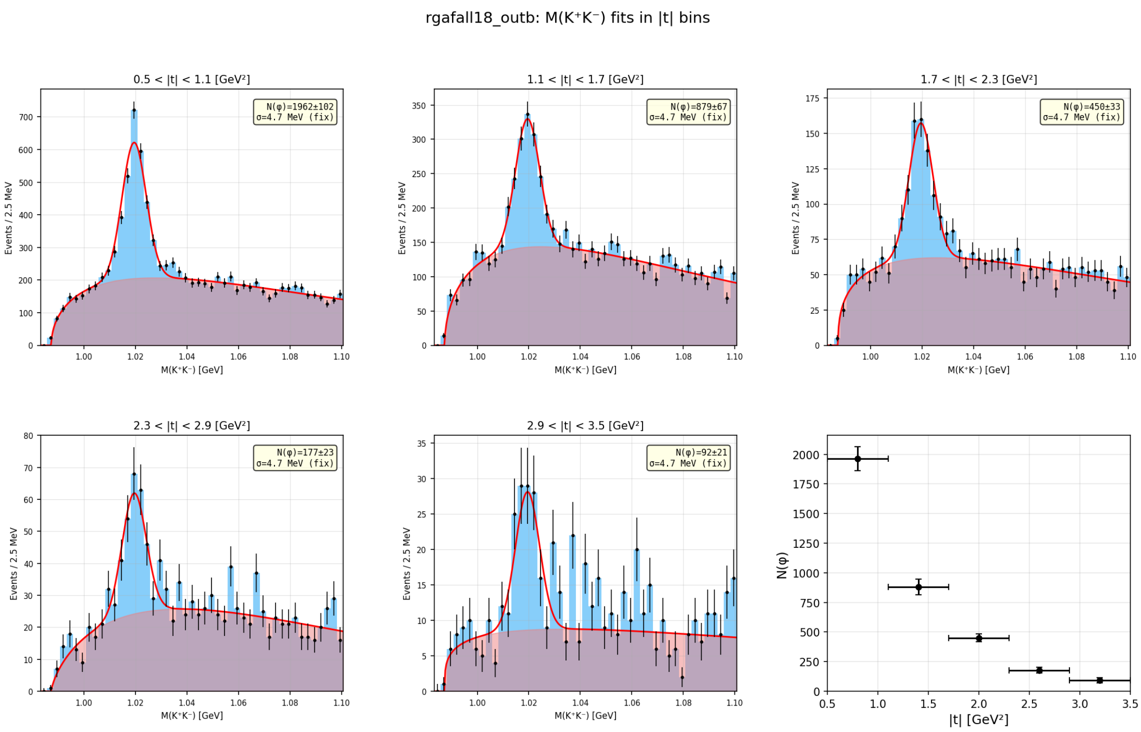

Figure 6: $M(K^+K^-)$ fits in bins of $|t|$ for RGA Fall 2018, outbending.

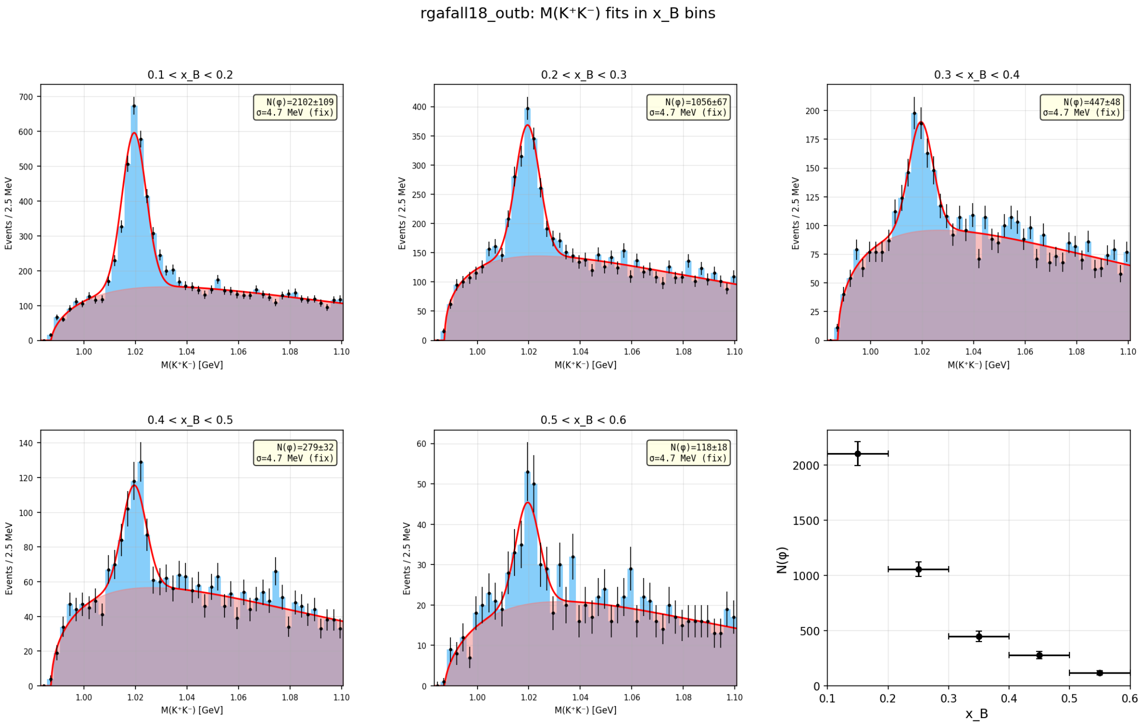

Figure 7: $M(K^+K^-)$ fits in bins of x_B for RGA Fall 2018, outbending.

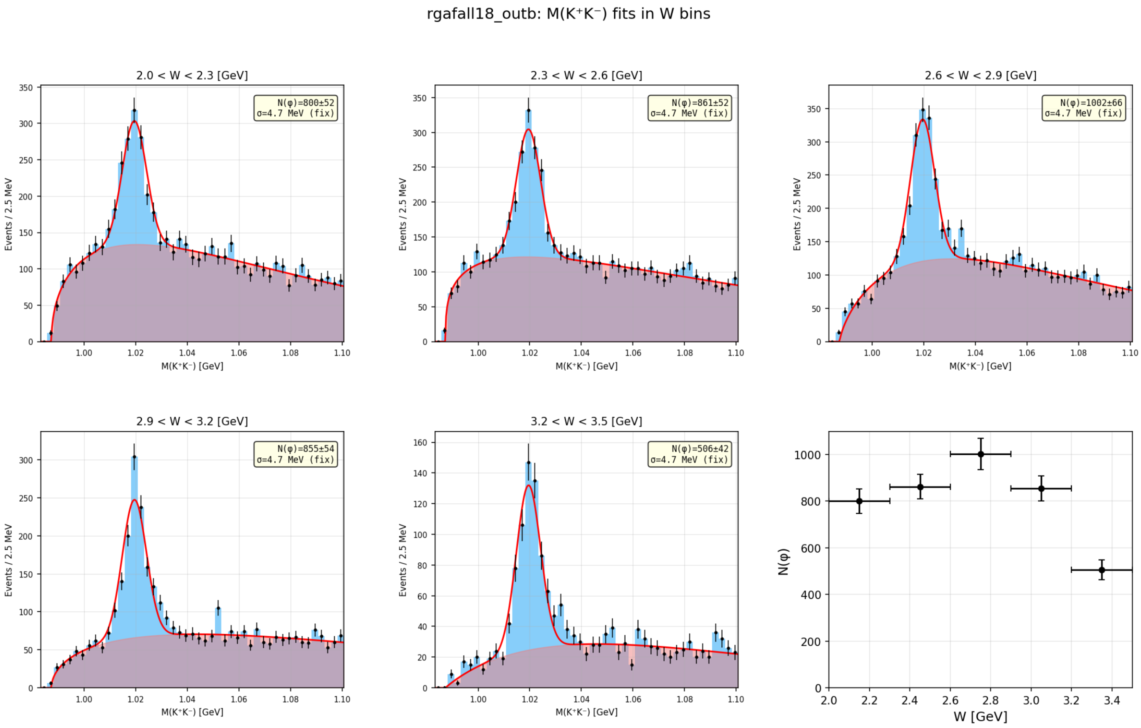

Figure 8: $M(K^+K^-)$ fits in bins of W for RGA Fall 2018, outbending.

RGA Spring 2018, inbending

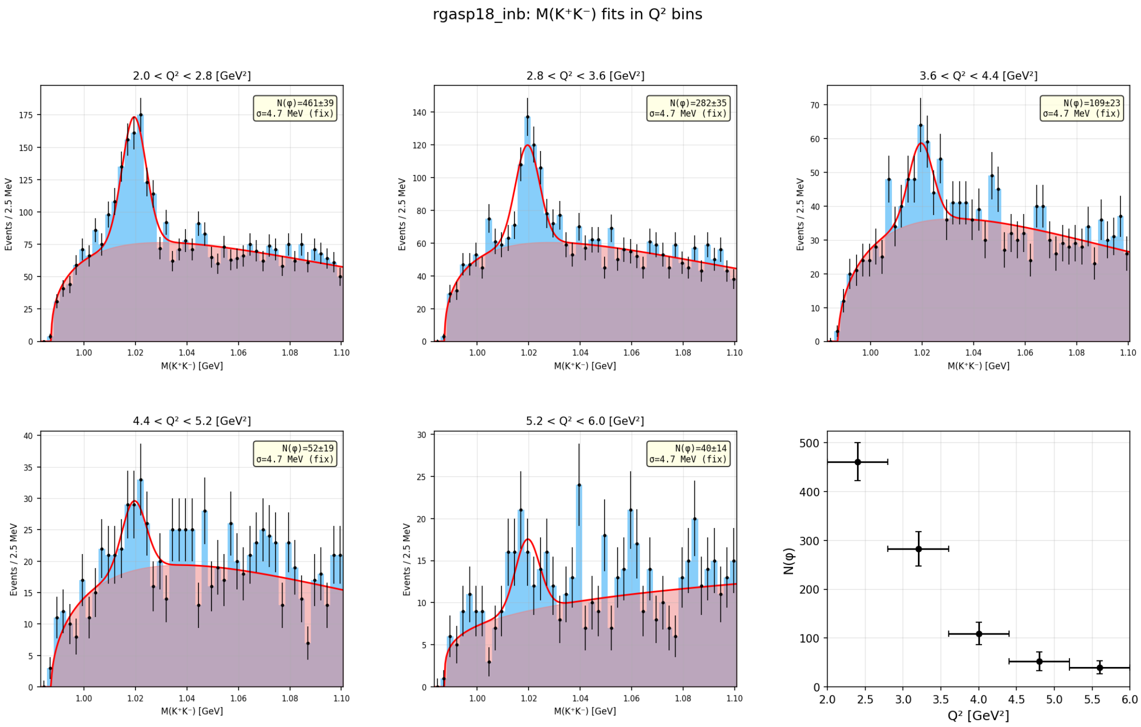

Figure 9: $M(K^+K^-)$ fits in bins of Q^2 for RGA Spring 2018, inbending.

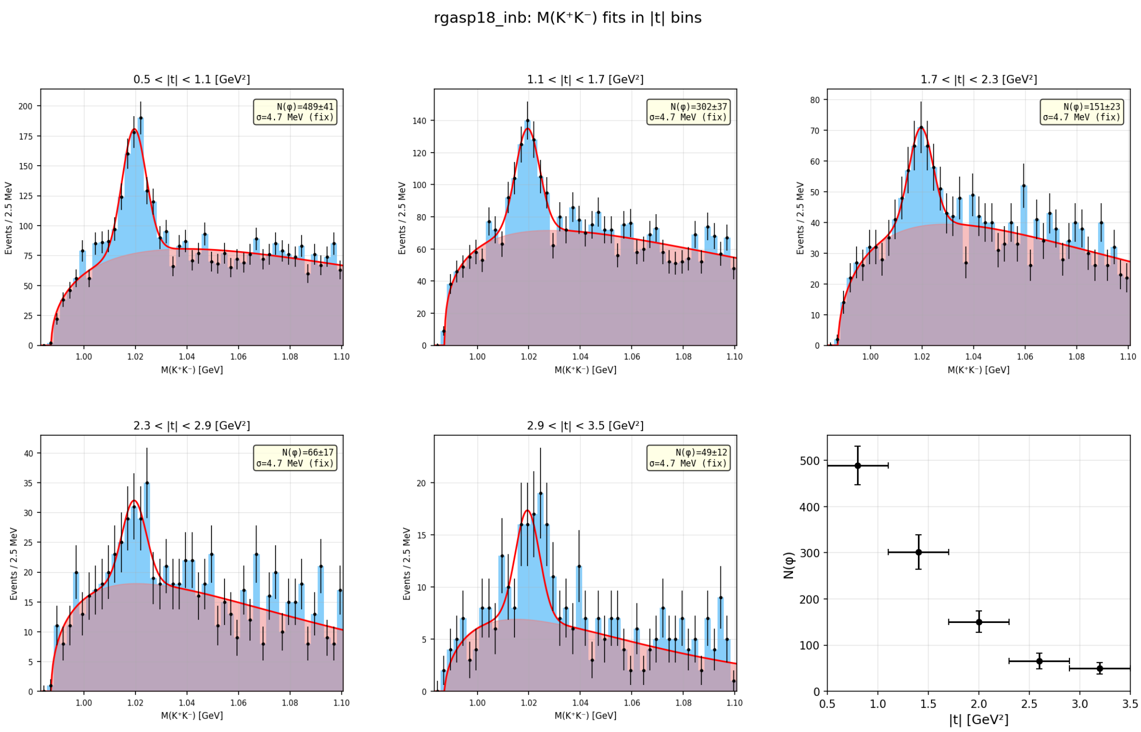

Figure 10: $M(K^+K^-)$ fits in bins of $|t|$ for RGA Spring 2018, inbending.

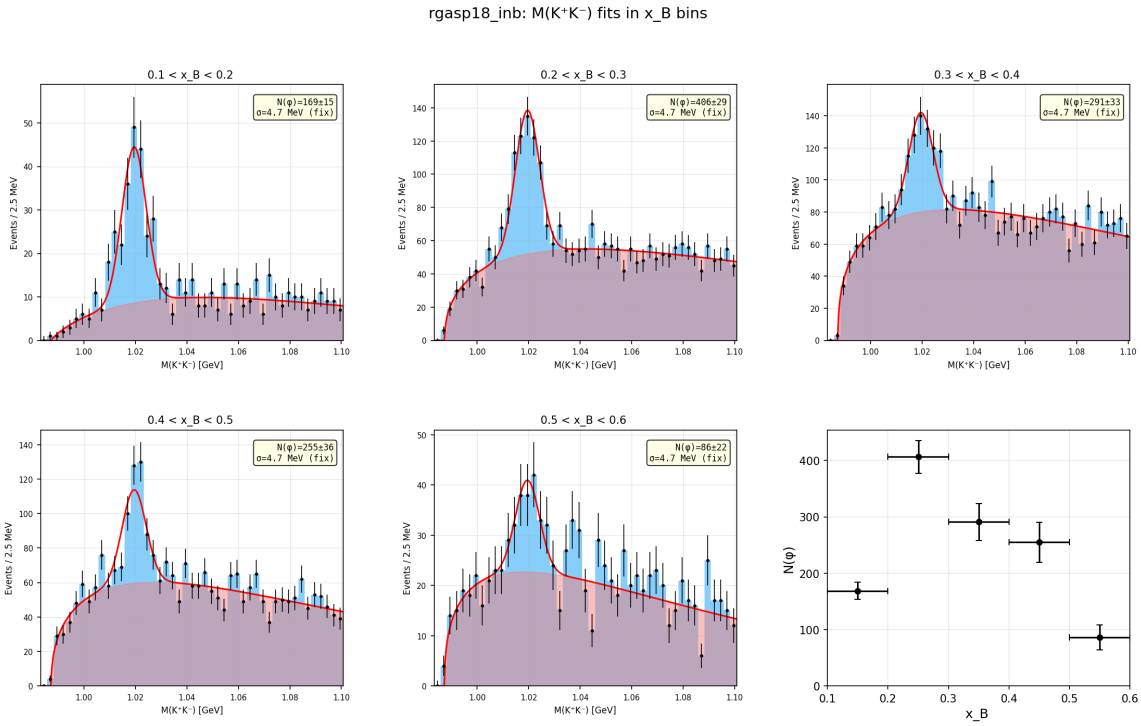

Figure 11: $M(K^+K^-)$ fits in bins of x_B for RGA Spring 2018, inbending.

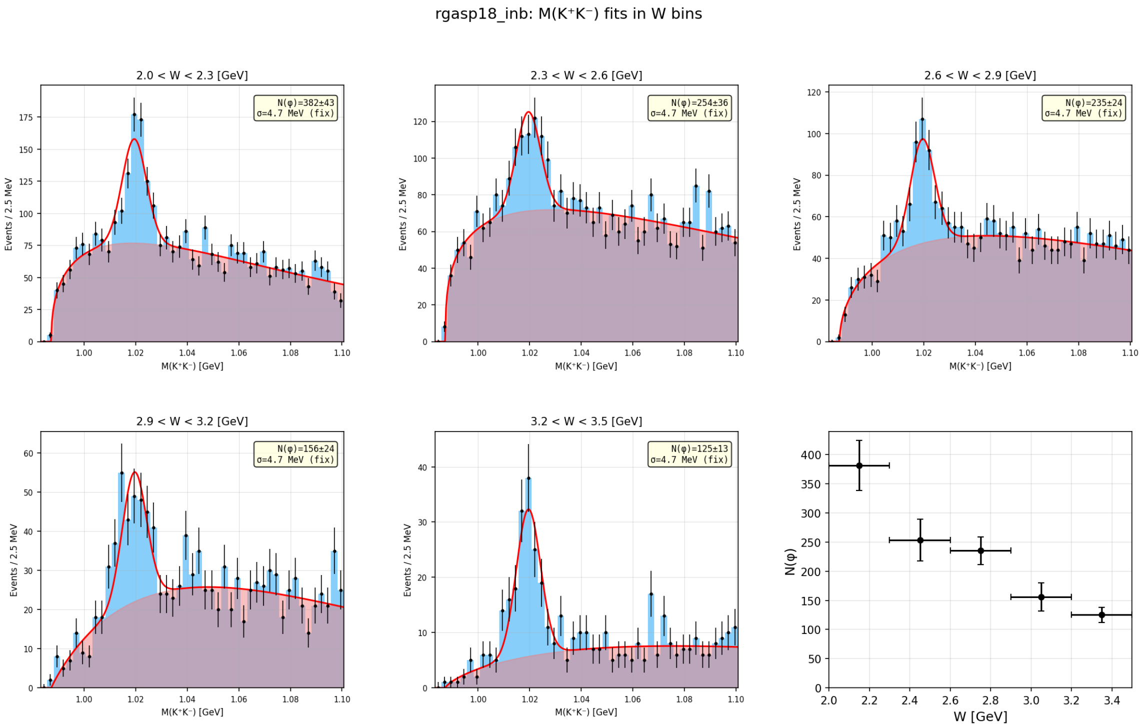

Figure 12: $M(K^+K^-)$ fits in bins of W for RGA Spring 2018, inbending.

RGA Spring 2018, outbending

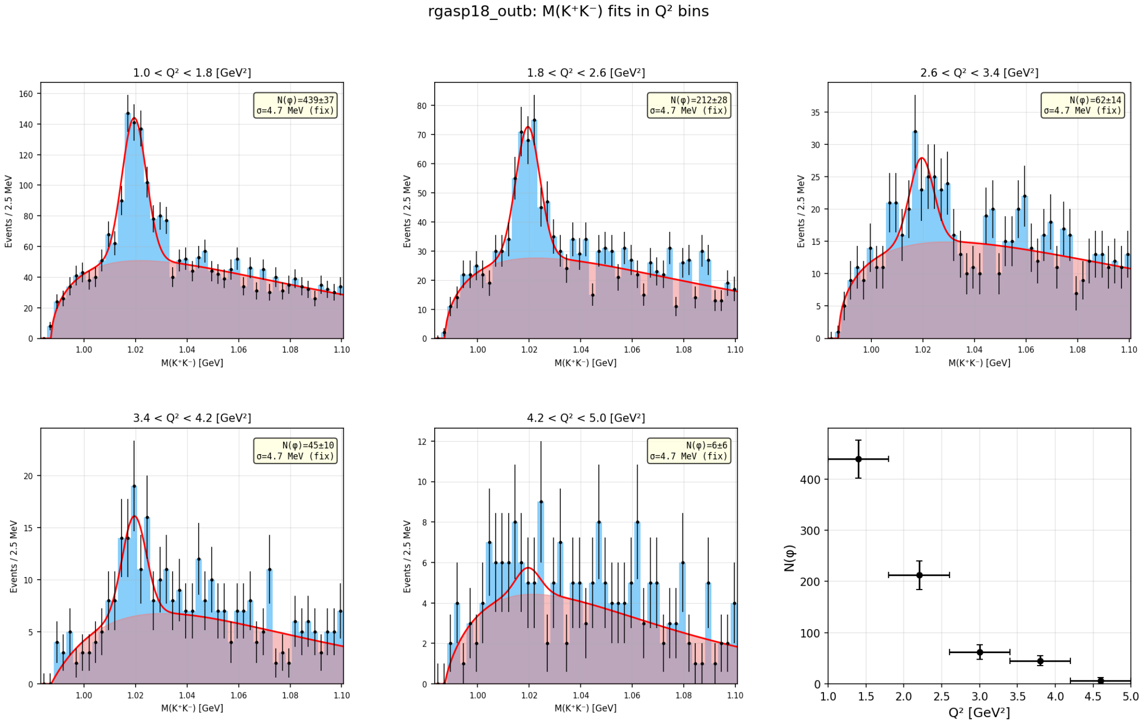

Figure 13: $M(K^+K^-)$ fits in bins of Q^2 for RGA Spring 2018, outbending.

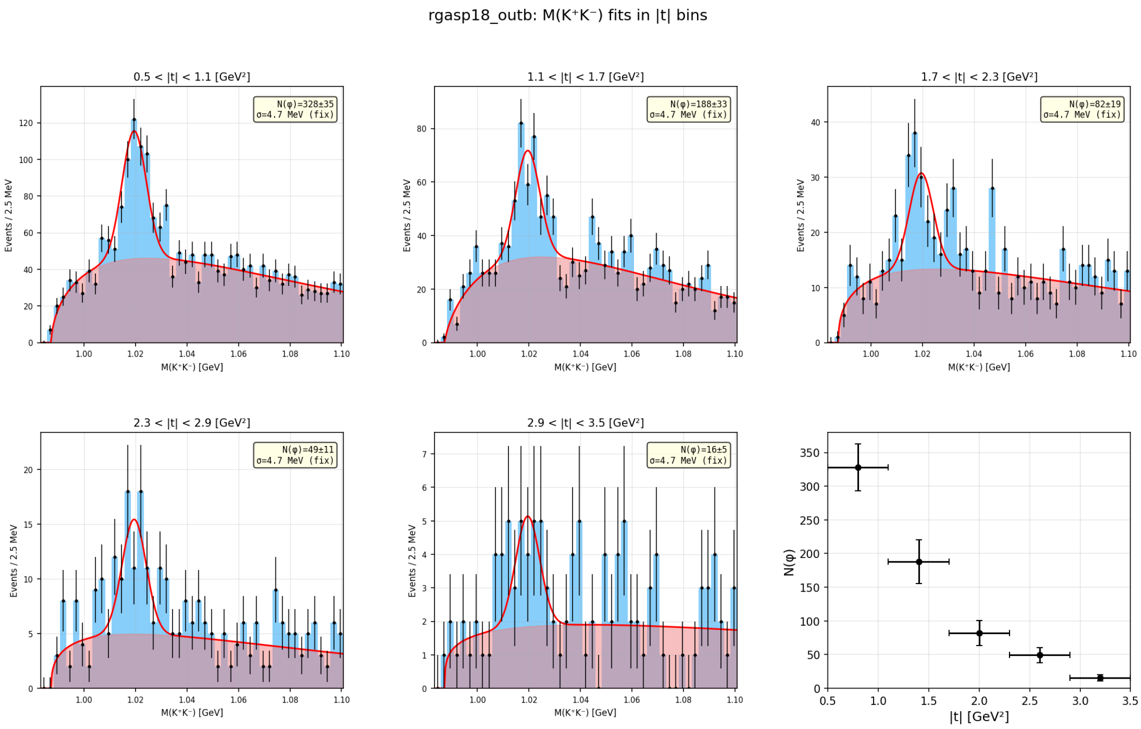

Figure 14: $M(K^+K^-)$ fits in bins of $|t|$ for RGA Spring 2018, outbending.

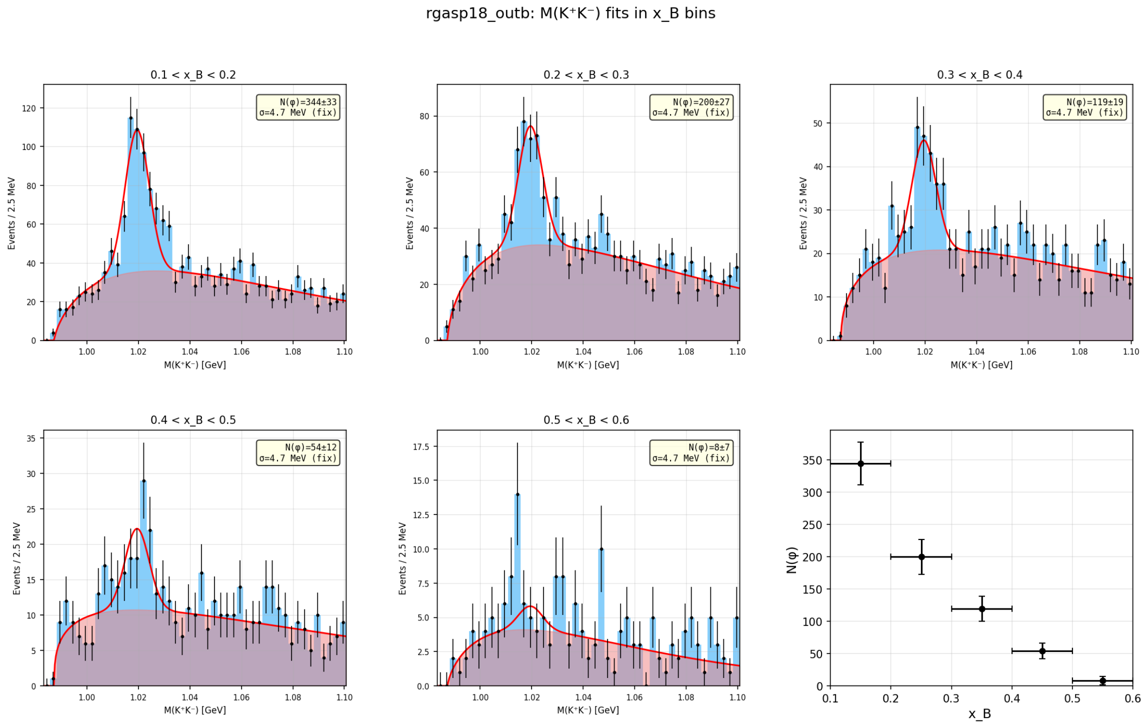

Figure 15: $M(K^+K^-)$ fits in bins of x_B for RGA Spring 2018, outbending.

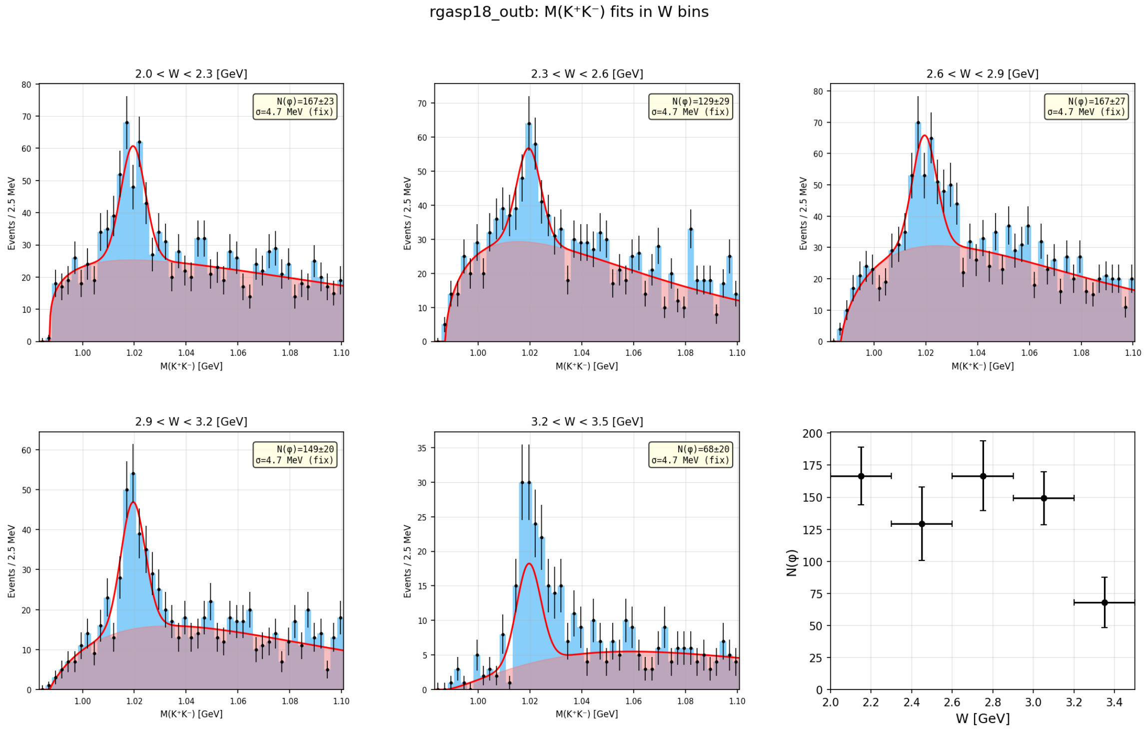

Figure 16: $M(K^+K^-)$ fits in bins of W for RGA Spring 2018, outbending.

RGA Spring 2019, inbending

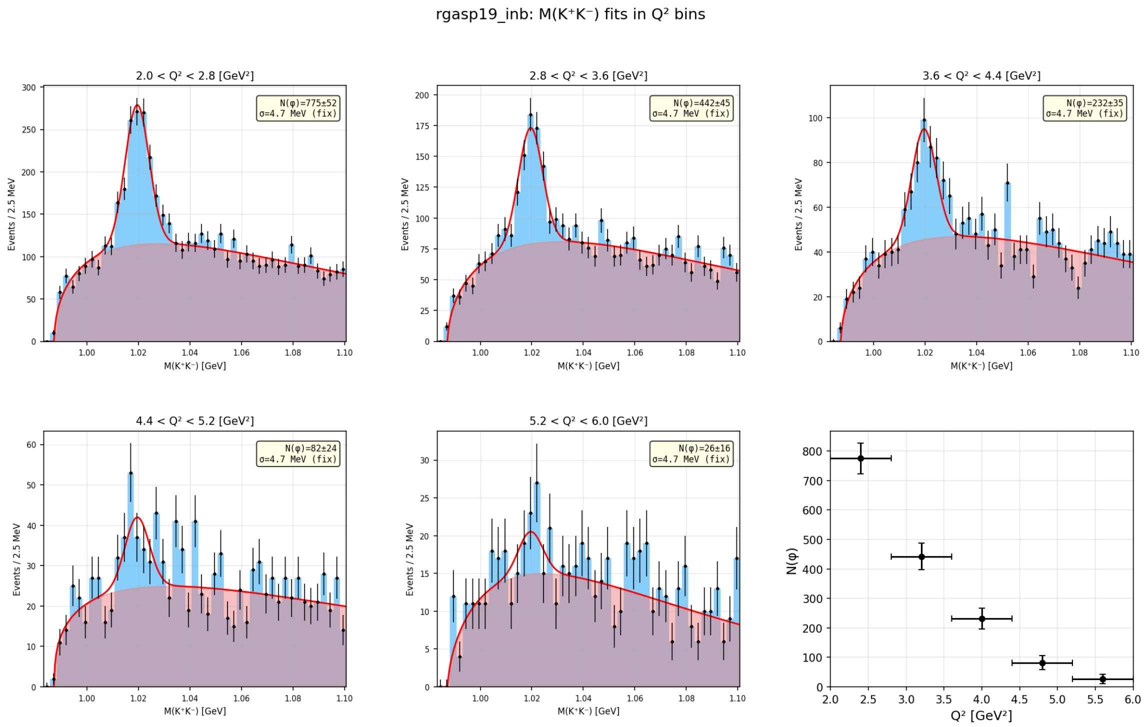

Figure 17: $M(K^+K^-)$ fits in bins of Q^2 for RGA Spring 2019, inbending.

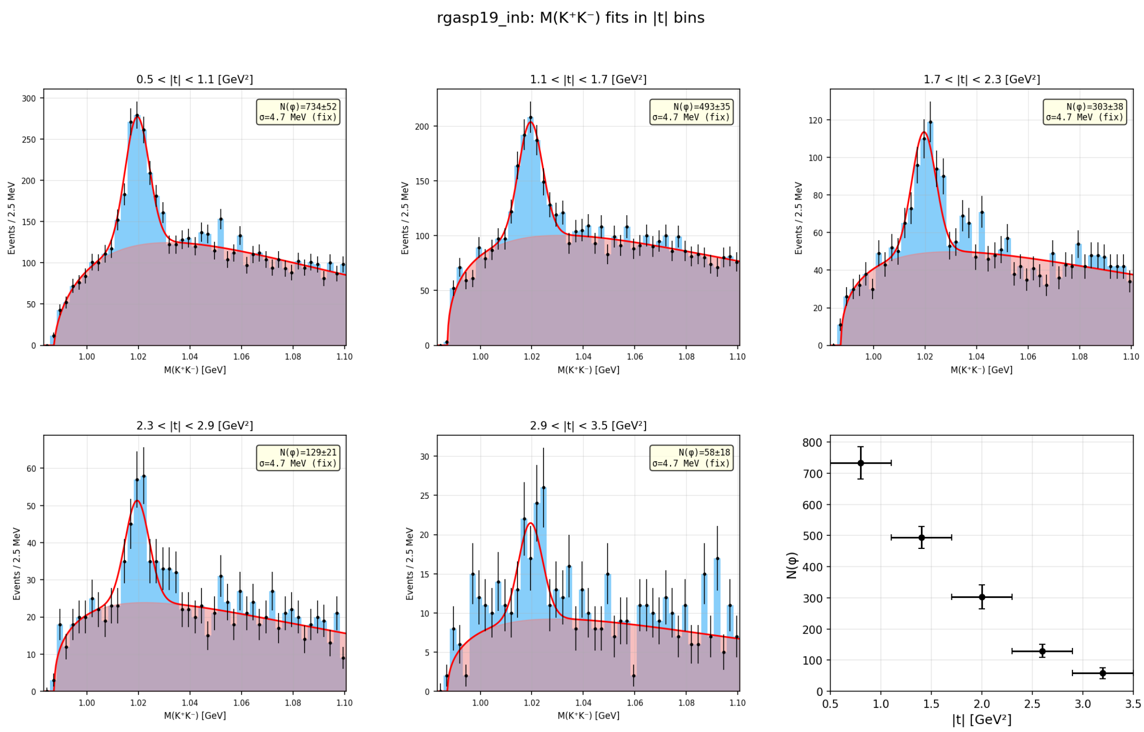

Figure 18: $M(K^+K^-)$ fits in bins of $|t|$ for RGA Spring 2019, inbending.

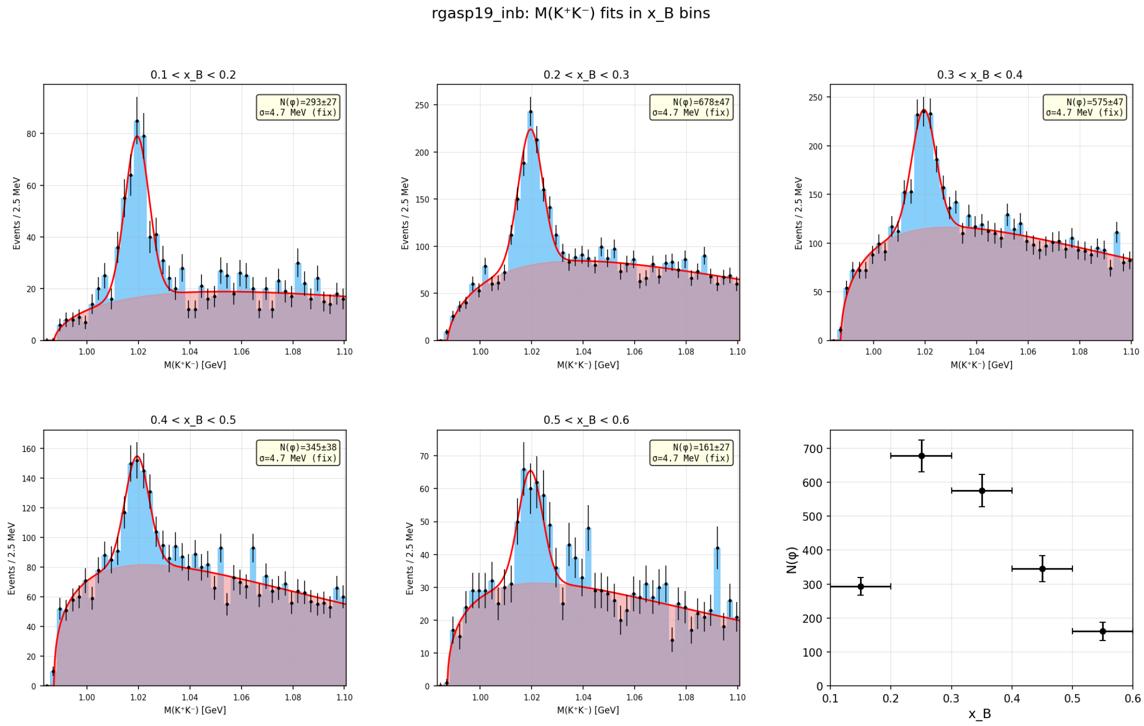

Figure 19: $M(K^+K^-)$ fits in bins of x_B for RGA Spring 2019, inbending.

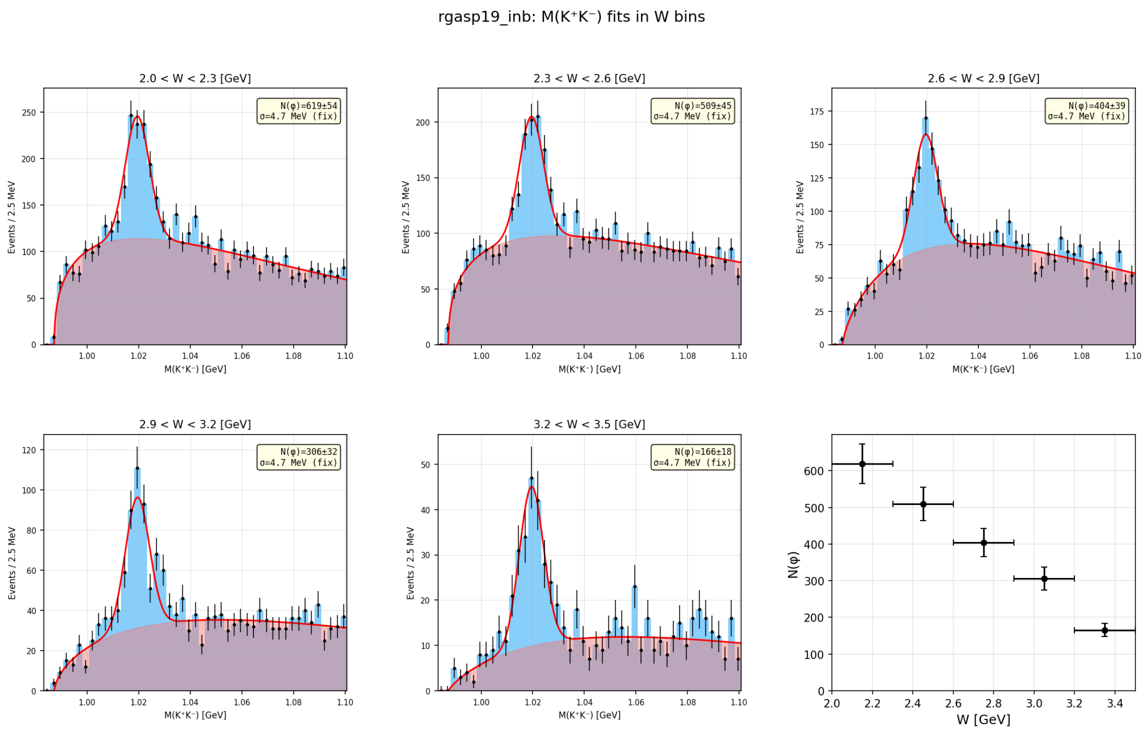

Figure 20: $M(K^+K^-)$ fits in bins of W for RGA Spring 2019, inbending.

5 Acceptance Correction

The acceptance ε_i in each kinematic bin i is computed from the Monte Carlo as:

$$\varepsilon_i = \frac{N_i^{\text{MC, accepted}}}{N_i^{\text{MC, generated}}} \tag{6}$$

where $N_i^{\text{MC, generated}}$ is the number of generated events in bin i after kinematic cuts, and $N_i^{\text{MC, accepted}}$ is the subset that passes the NN fast MC acceptance.

The acceptance-corrected yield is:

$$N_i^{\text{corr}} = \frac{N_i(\phi)}{\varepsilon_i} \tag{7}$$

where $N_i(\phi)$ is the ϕ signal yield from the $M(K^+K^-)$ fit.

5.1 Acceptance-Corrected Distributions with Model Fits

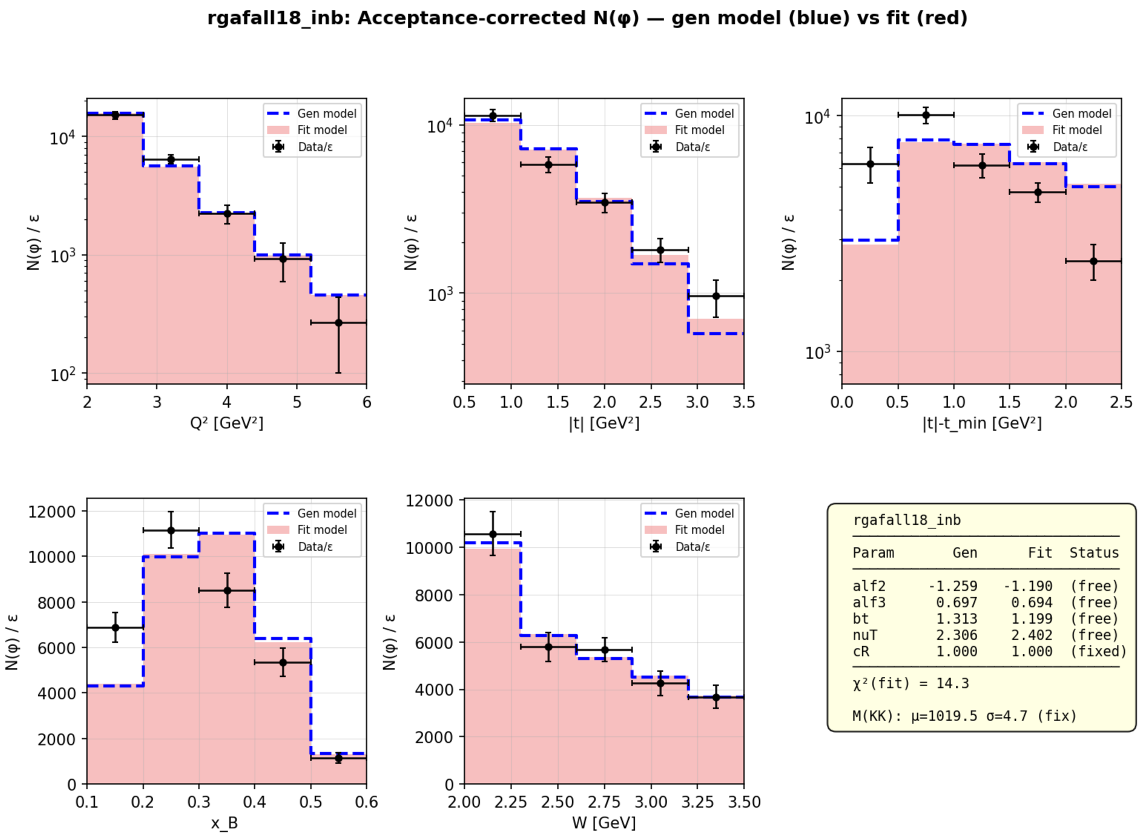

Figure 21: Acceptance-corrected distributions for RGA Fall 2018, inbending. Blue dashed histogram: generation model; red filled histogram: fitted model.

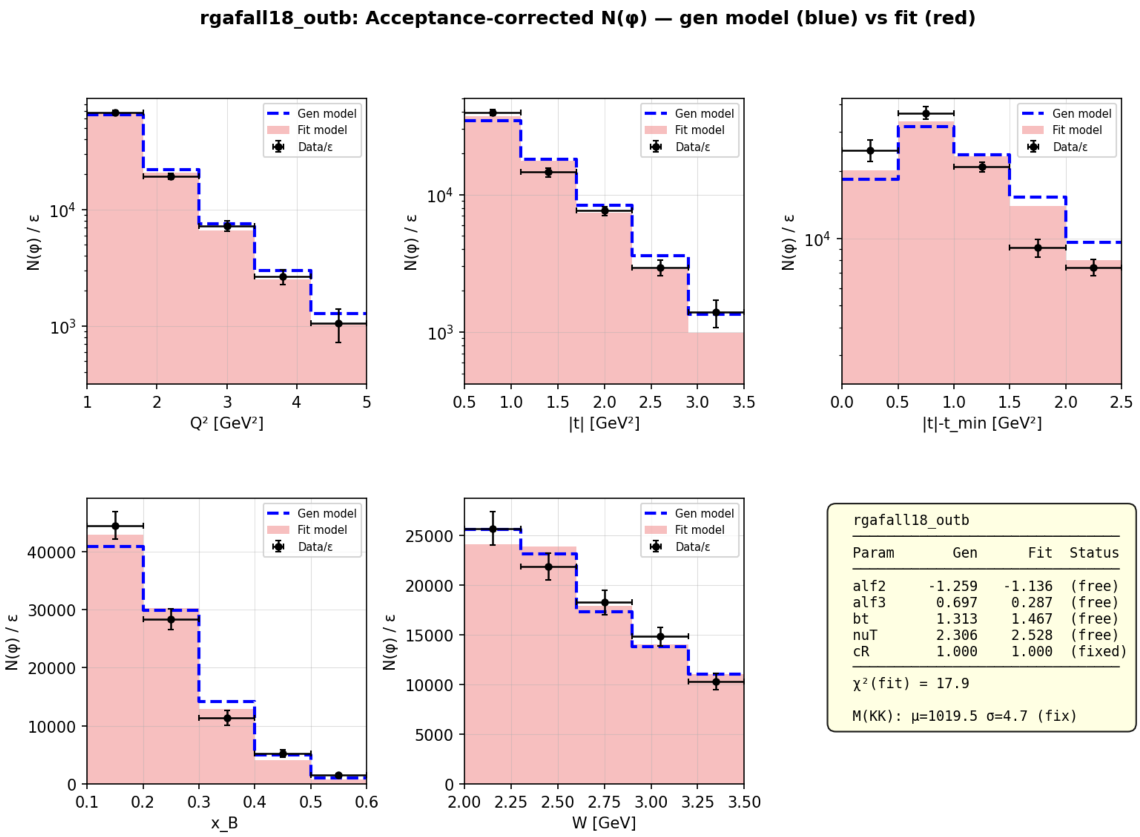

Figure 22: Acceptance-corrected distributions for RGA Fall 2018, outbending. Blue dashed histogram: generation model; red filled histogram: fitted model.

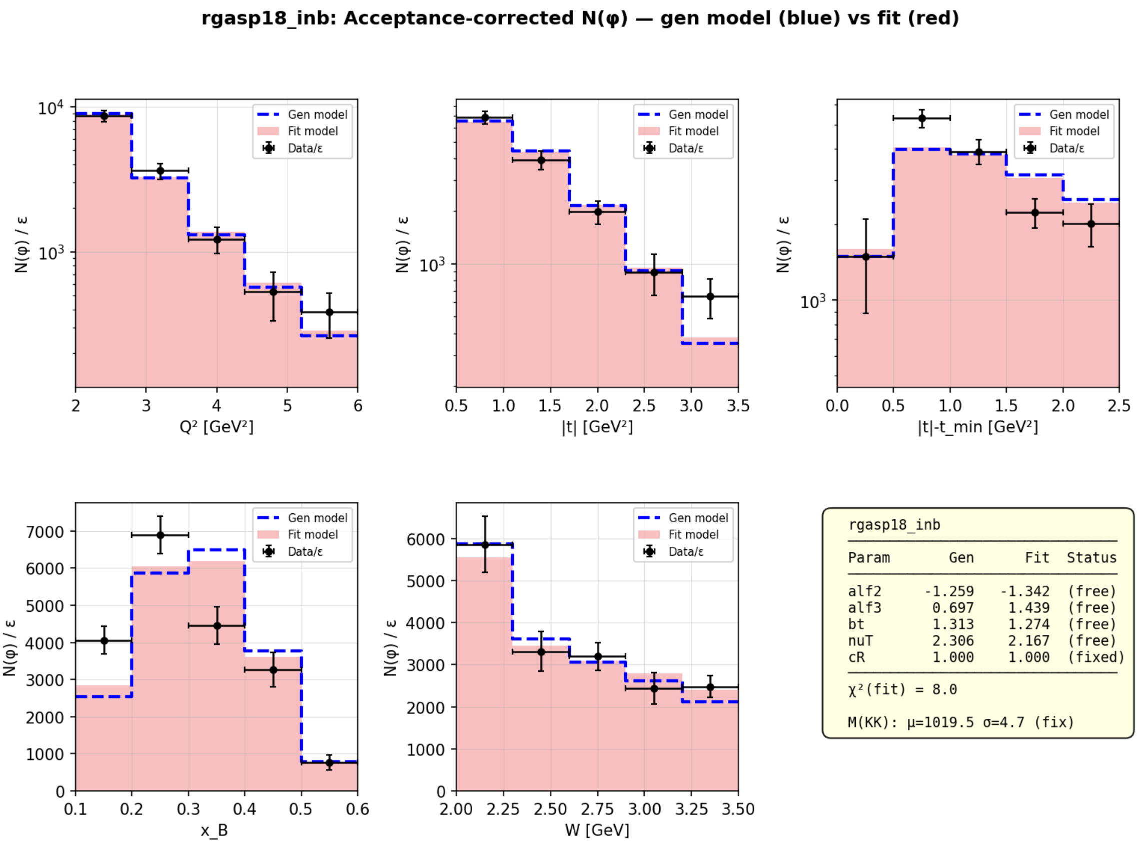

Figure 23: Acceptance-corrected distributions for RGA Spring 2018, inbending. Blue dashed histogram: generation model; red filled histogram: fitted model.

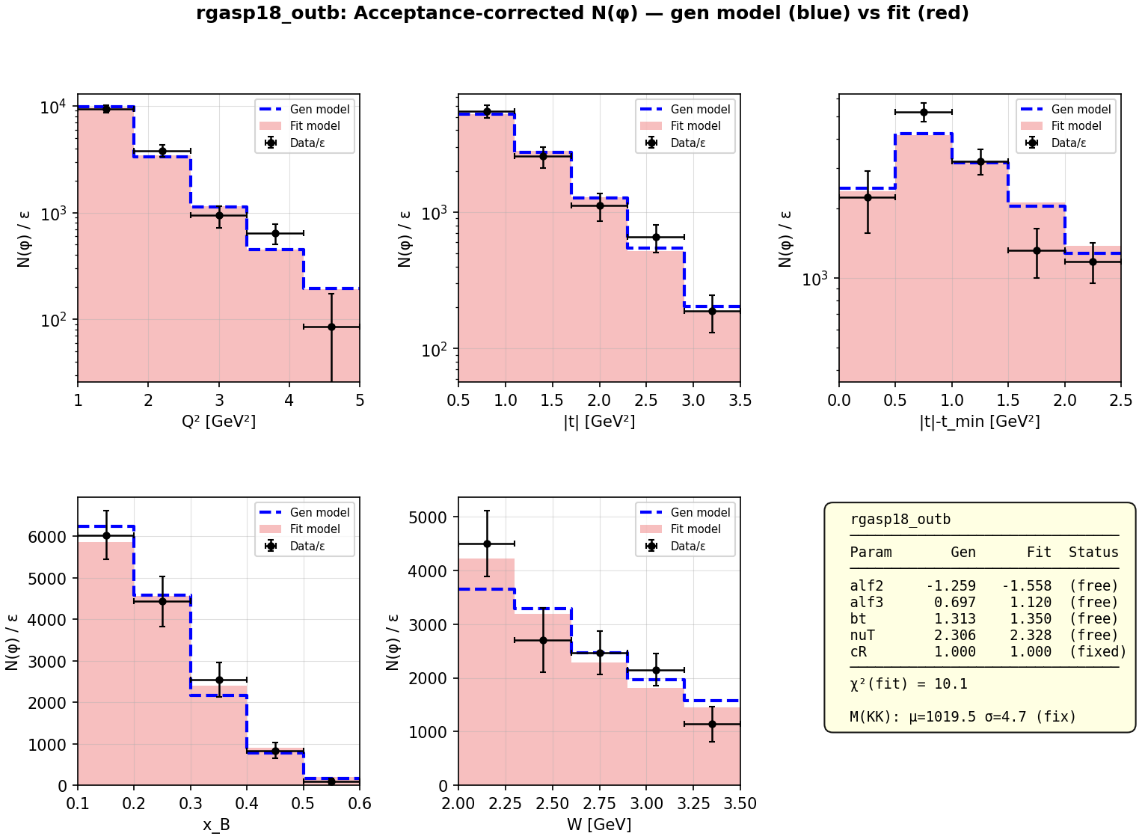

Figure 24: Acceptance-corrected distributions for RGA Spring 2018, outbending. Blue dashed histogram: generation model; red filled histogram: fitted model.

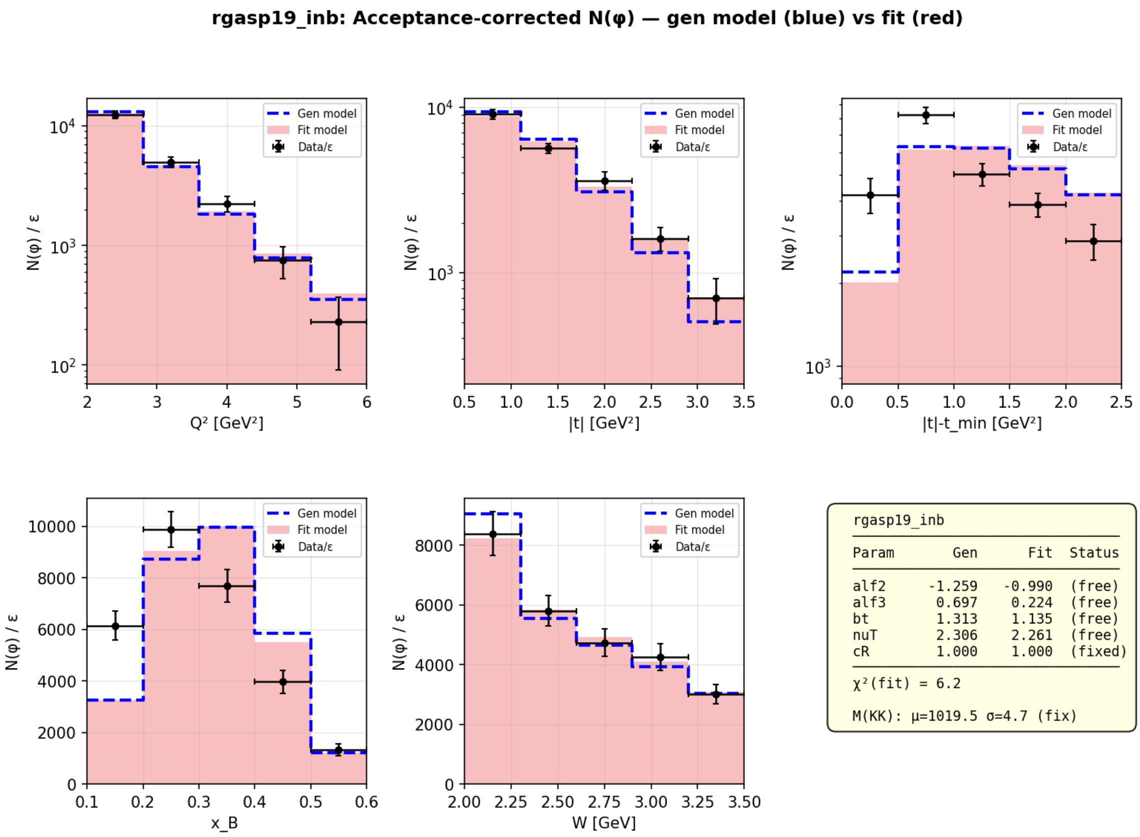

Figure 25: Acceptance-corrected distributions for RGA Spring 2019, inbending. Blue dashed histogram: generation model; red filled histogram: fitted model.

6 Model Fit Results

The cross-section model parameters are determined by minimizing:

$$\chi^2 = \sum_{\text{var}} \sum_i \frac{\left(N_i^{\text{corr}} - N_i^{\text{model}}\right)^2}{(\delta N_i^{\text{corr}})^2} \tag{8}$$

where the sum runs over the fitted kinematic variables (Q^2, $|t|$, W) and their bins (5 bins each, 15 data points total). The variables x_B and $|t| - |t_{\min}|$ are not fitted since the model has no independent parameters controlling them.

The model prediction N_i^{model} is obtained by reweighting the MC events:

$$N_i^{\text{model}} = C \sum_{j \in \text{bin}\, i} \frac{\sigma_{\text{new}}(Q_j^2, x_{B,j}, t_j)}{\sigma_{\text{gen}}(Q_j^2, x_{B,j}, t_j)} \tag{9}$$

where σ_{gen} is the cross section used in generation and σ_{new} uses the trial parameters. The normalization C is determined analytically.

The optimization uses `scipy.optimize.differential_evolution` with bounds: $\alpha_2 \in [-2, 5]$, $\alpha_3 \in [-15, 15]$, $b_t \in [0.1, 8]$, $\nu_T \in [0, 10]$. The parameter $c_R = 1$ is fixed.

6.1 Results

Table 1: Fitted model parameters for each dataset. Generation parameters: $\alpha_2 = -1.259$, $\alpha_3 = 0.697$, $b_t = 1.313$, $\nu_T = 2.306$. χ^2 is for 15 data points (5 bins $\times$ 3 variables) minus 4 free parameters.

Dataset	χ^2/ndf	α_2	α_3	b_t	ν_T
MC generation	—	-1.259	0.697	1.313	2.306
Fall18 inb	14.3/11	-1.190	0.694	1.199	2.402
Fall18 outb	17.9/11	-1.136	0.287	1.467	2.528
Sp18 inb	8.0/11	-1.342	1.439	1.274	2.167
Sp18 outb	10.1/11	-1.558	1.120	1.350	2.328
Sp19 inb	6.2/11	-0.990	0.224	1.135	2.261
Average	—	-1.243 ± 0.194	0.753 ± 0.470	1.285 ± 0.116	2.337 ± 0.123
Avg (inb)	—	-1.174 ± 0.144	0.786 ± 0.500	1.202 ± 0.057	2.277 ± 0.097
Avg (outb)	—	-1.347 ± 0.211	0.704 ± 0.416	1.409 ± 0.058	2.428 ± 0.100

Table 2: χ^2 per kinematic variable (5 bins each). Variables x_B and $|t| - |t_{\min}|$ are not included in the fit but shown for comparison.

Dataset	Q^2	$\|t\|$	W	x_B (not fit)	$\|t\| - \|t_{\min}\|$ (not fit)
Fall18 inb	4.0	8.3	2.0	29.8	76.8
Fall18 outb	2.5	10.3	4.6	19.0	44.6
Sp18 inb	1.8	4.5	1.4	25.7	17.0
Sp18 outb	4.8	1.7	3.3	2.6	12.2
Sp19 inb	3.1	2.6	0.4	53.4	61.7

6.2 Pipeline Fit Plots

7 Summary

The cross-section model for diffractive ϕ meson electroproduction has been tuned using five CLAS12 datasets with the generation parameters $\alpha_2 = -1.259$, $\alpha_3 = 0.697$, $b_t = 1.313$, $\nu_T = 2.306$.

The acceptance-corrected ϕ yields, extracted via $M(K^+K^-)$ fits with fixed Gaussian parameters ($\mu = 1019.5$ MeV, $\sigma = 4.7$ MeV), are fit using reweighting of the Monte Carlo events.

The average fitted parameters across all 5 datasets are:

Parameter	Average ± std
α_2	-1.243 ± 0.194
α_3	0.753 ± 0.470
b_t	1.285 ± 0.116
ν_T	2.337 ± 0.123

These values can be used as input for the next iteration of the generator.

A Generated Kinematics

B MC vs Data Comparison

C Acceptance Maps